What this subject is actually about

A plain-prose companion to `MBM_PRIMER.md`*. No equations to speak of. The primer has the machinery; this has the ideas. Read this first.*

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1. A number that isn't what it seems

A toxicologist puts liver cells in a plastic dish, adds a chemical at ten micromolar, waits a day, and writes down what happened. The dose, as reported in the paper, is ten micromolar.

The cells never saw ten micromolar.

From the moment the chemical enters the dish it starts going places. Some of it sticks to the plastic walls. Some dissolves into the fatty droplets that come with the serum in the growth medium. A great deal of it — for the chemicals we care about here, almost all of it — latches onto albumin, the carrier protein that serum is full of. Only what is left over, still dissolved in the water, is free to drift into a cell. By the time things settle down, the free concentration can be a thousandth of what was weighed out.

This matters enormously, because the reason anyone runs the experiment is to say something about people. The chain of reasoning goes: *this concentration harmed cells in a dish, therefore this blood level would be concerning in a person.* But a dish is not a bloodstream. A dish might be ten percent serum or half a percent; blood is blood. If you compare the number written on the dish with a number measured in blood, you are comparing two quantities that were never the same kind of thing.

The whole field exists to fix that comparison. Everything that follows is machinery in service of one question: **of everything you put in the well, how much actually reached the cell?**

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2. The idea: treat the dish as a budget

The fix is almost embarrassingly simple, and that simplicity is worth dwelling on, because the rest of the subject is just consequences of it.

Think of the well as a set of compartments: the water, the albumin, the fatty material, the cells themselves, the plastic. You know exactly how much chemical went in. None of it is destroyed — PFAS in particular are famously indestructible — and none escapes, because these molecules do not evaporate. So the amount you put in must still be there, distributed among the compartments. It is a budget, and budgets balance.

To work out the split you need only one thing per compartment: how strongly that compartment holds the chemical, compared with plain water. Give a number to each of those preferences, and the arithmetic of "the parts sum to the whole" does the rest. Out falls the free concentration in the water, and from that, everything else.

That is the entire model. One accounting identity and a list of preferences. When you read that two research groups have published competing mass-balance models, they are not disagreeing about the accounting. They are disagreeing about the list.

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3. The quantity that does all the work

The preference number has a name — a partition coefficient — and the easiest way to understand it is to picture the oldest experiment in the subject. Put some oil and some water in a jar. Add a pinch of your chemical. Shake it, let it settle into two layers, and measure the concentration in each. The ratio of the two is the partition coefficient. It says, in one number, how much this chemical would rather be in oil than in water.

Two features of that number matter here, and both come back to haunt us.

It is a ratio, not an amount. A chemical with a large partition coefficient for albumin is not necessarily mostly bound to albumin — that also depends on how much albumin is present. A strong preference for a phase that barely exists amounts to nothing. This sounds obvious written down, and it is the source of one of the two large errors we found in this project.

It is only a constant while the chemical is dilute. The underlying physics says the ratio holds when each molecule behaves as if it were alone, unaware of the others. Crowd enough of them in and they start to interfere — the oil gets saturated, the binding pockets on the protein fill up — and the "constant" quietly stops being constant. Nearly every subtlety in this subject is a version of that assumption failing.

4. Why PFAS break the mould

These models were built for a particular kind of chemical: neutral, greasy things like pesticides and industrial solvents, which partition into fat in a way that tracks hydrophobicity very well. For that family, one measurement predicts everything. PFAS are not in that family, for three reasons.

They are surfactants. A PFAS molecule is a water-repelling fluorinated tail attached to a water-loving acidic head — the same architecture as soap. Soap does not behave like oil. It collects at surfaces, forms films, and does things that a single "greasiness" number cannot anticipate.

They are charged at body pH. PFAS are strong acids, so in blood or in culture medium essentially every molecule carries a negative charge. Charge changes a molecule's preferences drastically — a charged species is far happier in water than its neutral twin. The models handle this by applying a penalty to the neutral prediction, but the size of that penalty is a modelling choice, and for albumin specifically, one of the models decides the penalty is *zero* — that the charged form binds albumin exactly as well as the neutral form would. That single choice is a large part of why albumin ends up dominating these calculations.

They bind albumin in a different way entirely. Most chemicals dissolve into a protein the way sugar dissolves into tea — diffusely, with no particular destination. PFAS dock into specific pockets on albumin, the same pockets that evolved to carry fatty acids. Pockets are countable, and countable things fill up. One of the two models is built entirely around that fact; the other ignores it.

5. Two philosophies

Set side by side, the two published models are a clean illustration of two ways to build any model at all.

Armitage's approach is to predict everything from one number. If you know how a chemical splits between oil and water, you can estimate how it splits between albumin and water, between cell membrane and water, between plastic and water — each by its own calibrated relationship. One input per chemical, and the whole well falls out. The appeal is enormous: it works for any chemical anyone has ever measured a greasiness number for, which is tens of thousands of them, and that is exactly what you need if your job is to screen a national inventory of industrial chemicals. It was built for breadth.

Qin's approach is to go to the bench. Measure how PFAS actually bind albumin, across a range of concentrations. Find — because this is what you do find — that the binding curve is not a straight line. At low concentrations each additional molecule binds as strongly as the last; at high concentrations the pockets are filling, and the next molecule has only weak, non-specific sticking available to it. Build that curve into the model so the albumin term weakens as you load it up. The result is far more faithful to what albumin does, but it exists only for the handful of chemicals that were measured. Everything else has to be guessed by interpolation. It was built for depth.

Neither choice is wrong. They are answers to different questions, and the tension between them runs through everything below.

6. What is a cell made of?

Here the two models part company completely, and it is worth seeing how far apart they get.

Armitage builds a cell as a recipe: so much storage fat, so much membrane fat, so much protein, the rest water, each ingredient contributing its own preference in proportion to how much of it there is. For a liver-derived cell line the recipe is overwhelmingly water — about ninety-four percent — with only a trace of fat and a few percent protein.

But the convention Armitage inherits says that most of a cell's protein does not sorb chemicals at all; only a small percentage of it behaves like a chemically interesting phase. Applied to PFAS, that assumption has a dramatic consequence. For a short-chain PFAS, which has little interest in fat, the model's cell is essentially a bag of water. For a long-chain one, which loves fat, the tiny membrane-fat fraction comes to dominate everything, and the model cell becomes, in effect, a droplet of lipid. The protein contributes a fraction of one percent throughout. **The convention has defined protein out of the cell.**

Qin's cell is the opposite. It is protein — over ninety percent of its sorptive capacity, at every chain length — with lipid playing a supporting role and no water phase at all. Where Armitage sees a water droplet turning into a fat droplet, Qin sees a lump of protein.

These are not small differences of emphasis. They are different pictures of what a cell is. And here is the thing that should make you uneasy: *the two models' final answers often agree anyway*. They agree because the albumin in the medium is so dominant that it swamps the cell term, and both models are really doing an albumin calculation with a cell attached. Agreement between two models is not corroboration of either when a third thing is doing the work.

7. Three experiments that do not agree with each other

Before you can say a model is wrong you need data to be wrong against. Three groups have put PFAS on liver cells and measured how much got inside.

They used different cells — two used a standard liver cancer line, the third used a more sophisticated line coaxed into behaving like real liver tissue. They used different amounts of serum, which as we have seen controls how much albumin is competing with the cells. They dissolved their chemicals in different solvents at wildly different strengths. They used different plate formats, so the ratio of plastic surface to liquid differs several-fold. One of them reported its results as a fraction of what was added; another as a fraction of what was *recovered*, which is not the same denominator.

And their results disagree. For one chemical at one dose, the third study reports roughly fourteen times more uptake than the other two. Even the two studies using the *same* cell line disagree by a factor of three or four on the short-chain compounds.

That is the honest starting point. A model that disagrees with these data by a factor of four is disagreeing with a target that is itself uncertain by a factor of four. Any conclusion has to survive that.

8. A sequence of disappointments

What happened when we tested the two models is a story worth telling in order, because the shape of it is more instructive than any individual result.

Both models over-predict. They both say more chemical should end up inside the cells than the experiments find. Not by a little.

So we looked for the wrong parameter. The obvious suspect was albumin, because albumin carries almost the entire budget — well over ninety percent for the longer chains. If one term holds ninety-five percent of the answer, an error there is an error in the answer. There is a very good modern measurement of PFAS–albumin binding, done at realistic concentrations, so we substituted it.

It made both models worse. The measurement says albumin binds PFAS *less* tightly than the models assume. A weaker albumin means less competition for the chemical, which means more of it free, which means more of it in the cell — and the models already predicted too much.

So we tried the cell instead. There is a measurement of how strongly PFAS bind cell protein, which is the term Armitage's convention had nearly erased. Substituting it made both models worse too — because that measurement says cell protein binds *more* than assumed, which enlarges the cell's appetite, which again sends more chemical inside.

Every correction pushed the same way. This is the pattern worth taking from the whole exercise: the best available measurement of each parameter moved the prediction further from the data, every time, because all the measurements point the same direction. You can be confident a parameter matters, confirm it is the most influential thing in your model, and still find it is not the thing that is wrong. **Sensitivity is not error.**

Then we found a mistake of our own. Reading the methods sections properly — which should have happened first — turned up two sentences saying that one of the three experiments had been run in a serum-*poor* medium, five times leaner than the growth medium quoted elsewhere in the paper. Our model had been giving that experiment five times more albumin than it had.

Fixing it made the fit *worse*, and that is the interesting part. The dataset that had looked like the models' best case was only looking good because of our error. Remove the error and it joins the others in disagreeing badly. A model that looks right for the wrong reason is more dangerous than one that looks wrong, because nobody investigates it.

Only one thing helps. The size of a cell. The published figures for how large a liver cell is span a factor of three, none of the three experiments measured their own, and adjusting it moves everything by roughly half the remaining gap. It is the one parameter that rescales every chemical by the same amount, which is what an unexplained offset needs. It is also the one parameter nobody bothered to measure, which tells you something about what the field thinks is interesting.

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9. The disagreement that depends on chain length

One pattern survived all of this and is worth understanding, because it says something structural rather than accidental.

PFAS come in families that differ by the length of their fluorinated tail. The longer the tail, the more the molecule loves fat and the more tightly it binds albumin. Armitage's over-prediction is not constant across that family — it grows steadily as the chains get longer, reaching about ten-fold for the longest. Qin's does not grow at all.

The explanation turns out to be elegant. Because albumin dominates the budget, what controls how much ends up in a cell is not the cell's appetite on its own, nor albumin's on its own, but the *ratio* between them. Whichever grows faster as you lengthen the chain wins.

Qin has that ratio roughly constant: his cell and his albumin grow at nearly the same rate, so the balance between them holds steady and there is no drift. Armitage has it growing, for two reasons that compound. His albumin relationship flattens out for very greasy chemicals — a deliberate feature, calibrated on ordinary drugs, where binding pockets do eventually stop rewarding extra greasiness. And his cell, being effectively a droplet of fat, keeps climbing steeply. A fat droplet gets more attractive with chain length faster than albumin does, so the ratio drifts, and the error grows.

So the two models fail in genuinely different ways. **Qin has the chemistry of chain length roughly right and the overall level wrong. Armitage has the level closer and the chain-length behaviour wrong.** They are not two attempts at the same thing with different accuracy; they are wrong about different things.

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10. The problem underneath all of it

Here is the thing that, once seen, reframes everything above.

How crowded is the albumin in these experiments? Each albumin molecule offers roughly one good binding site. In the dishes, there is somewhere between a third of a PFAS molecule and fourteen PFAS molecules for every albumin molecule. In human blood, there is about one PFAS molecule for every two thousand albumin molecules.

The experiments are running three to four orders of magnitude more crowded than the situation they are meant to inform. Their albumin is saturated; the body's albumin is essentially empty. And as we saw in section 3, a partition coefficient is only a constant while things are uncrowded. In these dishes it is not a constant at all — over a single dose series the apparent strength of albumin binding changes sevenfold.

This is why Qin's model needed its extra machinery: he measured the curve, saw it bend, and built the bending in. Armitage's model assumes a straight line, which is defensible for the dilute neutral chemicals it was designed for and is simply not true here.

But the deeper implication is uncomfortable, and it is the honest conclusion of this whole line of work. Even a *perfect* model of these dishes would be describing a chemical regime that does not occur in a person. Better bookkeeping inside the well gives you a better description of the experiment. It does not obviously give you a better answer about human exposure, because the experiment is in the wrong regime to begin with.

11. The same idea, somewhere it works

It would be easy to conclude from all this that the mass-balance idea is unsound. It isn't, and there is a clean demonstration.

Take the identical reasoning to blood serum. Serum is a simpler system than a culture well: water, albumin, and the globulins, with no plastic and no cells. One group measured, separately and by the same method, how strongly two dozen PFAS bind to purified albumin, how strongly they bind to purified globulin, and how strongly they bind to whole serum. That is exactly the test: predict the third from the first two, with nothing fitted.

It works. The predictions sit on the measurements with no systematic bias and a typical error of about fifty percent — which, for a quantity spanning a hundred-fold across the compound set, is good. Letting the protein concentrations float to improve the fit returns the physiological values and improves nothing, which is the strongest evidence of all that the structure is right rather than being propped up by its parameters.

So the accounting is sound. Free concentration times a sum of appetites really does describe a protein-rich fluid. What fails is the *dish* — a system with more phases, more uncertain parameters, a plastic term nobody can pin down, cells whose size nobody measured, and a crowding regime where the central assumption does not hold.

12. What the subject teaches

Strip away the chemistry and several lessons generalise well beyond PFAS.

A model is an accounting identity plus a list of preferences. The identity is never in doubt. All the argument is about the list, and about whether each preference was measured in the regime where it is being used.

Sensitivity is not error. Albumin controls almost the entire answer in these wells, which makes it the obvious suspect and makes it tempting to tune. It is not where the error is. The two are different questions and conflating them wastes a great deal of time.

An error that scales everything cancels; an error that changes a ratio survives. The three labs used solvents differing thirty-fold in strength, and it hardly matters, because a solvent weakens every preference by about the same factor and the model's answer depends on their ratios. The serum error, which changed how much albumin there was *relative to* everything else, was worth a factor of four. Knowing which kind of error you are looking at tells you immediately whether to care.

Agreement is not corroboration. The two models hold opposite views of what a cell is and frequently agree on the answer, because a third term dominates both. Two methods agreeing means something only if they could have disagreed.

The methods section is not optional. Two sentences about serum content, buried in a paragraph about cell culture, were worth more than every parameter substitution we tried. Nobody writes papers about reading the methods section carefully, and it was the highest-yield thing done in this project.

Know which regime your measurement came from. A binding constant measured on saturated protein and a binding constant measured on empty protein are different numbers for the same quantity, differing by more than an order of magnitude. The literature mostly does not record which was which, which makes a great deal of it unusable without detective work.

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13. When you want the machinery

Everything asserted here is derived, with numbers, in the companion documents:

- `MBM_PRIMER.md` — the equations, the approximations named one by one, the derivations, and worked numerical checks. Read it when you want to see *why* something above is true rather than take it on trust.
- `MBM_Qin_vs_Armitage.md` — the actual comparison against the three datasets, the medium correction, and the chain-length analysis.
- `ALBUMIN_BINDING.md` — albumin binding in its own right, including the serum result of section 11.